background in linear systems theory can skip the material in this chapter. However, it would still help to at least review this chapter to solidify the foundational concepts of state estimation before moving on to the later chapters of this book.

1.1 MATRIX ALGEBRA AND MATRIX CALCULUS

In this section, we review matrices, matrix algebra, and matrix calculus. This is necessary in order to understand the rest of the book because optimal state estimation algorithms are usually formulated with matrices.

A scalar is a single quantity. For example, the number 2 is a scalar. The number $1 + 3j$ is a scalar (we use j in this book to denote the square root of -1). The number π is a scalar.

A vector consists of scalars that are arranged in a row or column. For example, the vector

$$\begin{bmatrix} 1 & 3 & \pi \end{bmatrix} \tag{1.1}$$

is a 3-element vector. This vector is a called a 1×3 vector because it has 1 row and 3 columns. This vector is also called a row vector because it is arranged as a single row. The vector

$$\begin{bmatrix} -2 \\ \pi^2 \\ j \\ 0 \end{bmatrix} \tag{1.2}$$

is a 4-element vector. This vector is a called a 4×1 vector because it has 4 rows and 1 column. This vector is also called a column vector because it is arranged as a single column. Note that a scalar can be viewed as a 1-element vector; a scalar is a degenerate vector. (This is just like a plane can be viewed as a 3-dimensional shape; a plane is a degenerate 3-dimensional shape.)

A matrix consists of scalars that are arranged in a rectangle. For example, the matrix

$$\begin{bmatrix} -2 & 3 \\ 0 & \pi^2 \\ j & 0 \end{bmatrix} \tag{1.3}$$

is a 3×2 matrix because it has 3 rows and 2 columns. The number of rows and columns in a matrix can be collectively referred to as the dimension of the matrix. For example, the dimension of the matrix in the preceding equation is 3×2. Note that a vector can be viewed as a degenerate matrix. For example, Equation (1.1) is a 1×3 matrix. A scalar can also be viewed as a degenerate matrix. For example, the scalar 6 is a 1×1 matrix.

The rank of a matrix is defined as the number of linearly independent rows. This is also equal to the number of linearly independent columns. The rank of a matrix A is often indicated with the notation $\rho(A)$. The rank of a matrix is always less than or equal to the number of rows, and it is also less than or equal to the number of columns. For example, the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \tag{1.4}$$

has a rank of one because it has only one linearly independent row; the two rows are multiples of each other. It also has only one linearly independent column; the two columns are multiples of each other. On the other hand, the matrix

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \tag{1.5}$$

has a rank of two because it has two linearly independent rows. That is, there are no nonzero scalars c_1 and c_2 such that

$$c_1 \begin{bmatrix} 1 & 3 \end{bmatrix} + c_2 \begin{bmatrix} 2 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix} \tag{1.6}$$

so the two rows are linearly independent. It also has two linearly independent columns. That is, there are no nonzero scalars c_1 and c_2 such that

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{1.7}$$

so the two columns are linearly independent. A matrix whose elements are comprised entirely of zeros has a rank of zero. An $n \times m$ matrix whose rank is equal to $\min(n,m)$ is called full rank. The nullity of an $n \times m$ matrix A is equal to $[m - \rho(A)]$.

The transpose of a matrix (or vector) can be taken by changing all the rows to columns, and all the columns to rows. The transpose of a matrix is indicated with a T superscript, as in A^T.[1] For example, if A is the $r \times n$ matrix

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & & \vdots \\ A_{r1} & \cdots & A_{rn} \end{bmatrix} \tag{1.8}$$

then A^T is the $n \times r$ matrix

$$A^T = \begin{bmatrix} A_{11} & \cdots & A_{r1} \\ \vdots & & \vdots \\ A_{1n} & \cdots & A_{rn} \end{bmatrix} \tag{1.9}$$

Note that we use the notation A_{ij} to indicate the scalar in the ith row and jth column of the matrix A. A symmetric matrix is one for which $A = A^T$.

The hermitian transpose of a matrix (or vector) is the complex conjugate of the transpose, and is indicated with an H superscript, as in A^H. For example, if

$$A = \begin{bmatrix} 1 & 2j & 3-j \\ 4j & 5+j & 1-3j \end{bmatrix} \tag{1.10}$$

then

$$A = \begin{bmatrix} 1 & -4j \\ -2j & 5-j \\ 3+j & \\ 1+3j & \end{bmatrix} \tag{1.11}$$

A hermitian matrix is one for which $A = A^H$.

[1] Many papers or books indicate transpose with a prime, as in A', or with a lower case t, as in A^t.

1.1.1 Matrix algebra

Matrix addition and subtraction is simply defined as element-by-element addition and subtraction. For example,

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 4 & 1 \\ 1 & -1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 6 & 4 \\ 4 & 1 & -1 \end{bmatrix} \tag{1.12}$$

The sum $(A + B)$ and the difference $(A - B)$ is defined only if the dimension of A is equal to the dimension of B.

Suppose that A is an $n \times r$ matrix and B is an $r \times p$ matrix. Then the product of A and B is written as $C = AB$. Each element in the matrix product C is computed as

$$C_{ij} = \sum_{k=1}^{r} A_{ik} B_{kj} \qquad i = 1, \ldots, n \qquad j = 1, \ldots, p \tag{1.13}$$

The matrix product AB is defined only if the number of columns in A is equal to the number of rows in B. It is important to note that matrix multiplication does not commute. In general, $AB \neq BA$.

Suppose we have an $n \times 1$ vector x. We can compute the 1×1 product $x^T x$, and the $n \times n$ product xx^T as follows:

$$\begin{aligned} x^T x &= \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\ &= x_1^2 + \cdots + x_n^2 \\ xx^T &= \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \\ &= \begin{bmatrix} x_1^2 & \cdots & x_1 x_n \\ \vdots & \ddots & \vdots \\ x_n x_1 & \cdots & x_n^2 \end{bmatrix} \end{aligned} \tag{1.14}$$

Suppose that we have a $p \times n$ matrix H and an $n \times n$ matrix P. Then H^T is a $n \times p$ matrix, and we can compute the $p \times p$ matrix product HPH^T.

$$\begin{aligned} HPH^T &= \begin{bmatrix} H_{11} & \cdots & H_{1n} \\ \vdots & \ddots & \vdots \\ H_{p1} & \cdots & H_{pn} \end{bmatrix} \begin{bmatrix} P_{11} & \cdots & P_{1n} \\ \vdots & \ddots & \vdots \\ P_{n1} & \cdots & P_{nn} \end{bmatrix} \begin{bmatrix} H_{11} & \cdots & H_{p1} \\ \vdots & \ddots & \vdots \\ H_{1n} & \cdots & H_{pn} \end{bmatrix} \\ &= \begin{bmatrix} \sum_{j,k} H_{1j} P_{jk} H_{1k} & \cdots & \sum_{j,k} H_{1j} P_{jk} H_{pk} \\ \vdots & \ddots & \vdots \\ \sum_{j,k} H_{pj} P_{jk} H_{1k} & \cdots & \sum_{j,k} H_{pj} P_{jk} H_{pk} \end{bmatrix} \end{aligned} \tag{1.15}$$

This matrix of sums can be written as the following sum of matrices:

$$\begin{aligned}
HPH^T &= \begin{bmatrix} H_{11}P_{11}H_{11} & \cdots & H_{11}P_{11}H_{p1} \\ \vdots & \ddots & \vdots \\ H_{p1}P_{11}H_{11} & \cdots & H_{p1}P_{11}H_{p1} \end{bmatrix} + \cdots + \\
&\quad \begin{bmatrix} H_{1n}P_{nn}H_{1n} & \cdots & H_{1n}P_{nn}H_{pn} \\ \vdots & \ddots & \vdots \\ H_{pn}P_{nn}H_{1n} & \cdots & H_{pn}P_{nn}H_{pn} \end{bmatrix} \\
&= H_1P_{11}H_1^T + \cdots + H_nP_{nn}H_n^T \\
&= \sum_{j,k} H_jP_{jk}H_k^T \qquad (1.16)
\end{aligned}$$

where we have used the notation that H_k is the kth column of H.

Matrix division is not defined; we cannot divide a matrix by another matrix (unless, of course, the denominator matrix is a scalar).

An identity matrix I is defined as a square matrix with ones on the diagonal and zeros everywhere else. For example, the 3×3 identity matrix is equal to

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad (1.17)$$

The identity matrix has the property that $AI = A$ for any matrix A, and $IA = A$ (as long the dimensions of the identity matrices are compatible with those of A). The 1×1 identity matrix is equal to the scalar 1.

The determinant of a matrix is defined inductively for square matrices. The determinant of a scalar (i.e., a 1×1 matrix) is equal to the scalar. Now consider an $n \times n$ matrix A. Use the notation $A^{(i,j)}$ to denote the matrix that is formed by deleting the ith row and jth column of A. The determinant of A is defined as

$$|A| = \sum_{j=1}^{n} (-1)^{i+j} A_{ij} |A^{(i,j)}| \qquad (1.18)$$

for any value of $i \in [1, n]$. This is called the Laplace expansion of A along its ith row. We see that the determinant of the $n \times n$ matrix A is defined in terms of the determinants of $(n-1) \times (n-1)$ matrices. Similarly, the determinants of $(n-1) \times (n-1)$ matrices are defined in terms of the determinants of $(n-2) \times (n-2)$ matrices. This continues until the determinants of 2×2 matrices are defined in terms of the determinants of 1×1 matrices, which are scalars. The determinant of A can also be defined as

$$|A| = \sum_{i=1}^{n} (-1)^{i+j} A_{ij} |A^{(i,j)}| \qquad (1.19)$$

for any value of $j \in [1, n]$. This is called the Laplace expansion of A along its jth column. Interestingly, Equation (1.18) (for any value of i) and Equation (1.19) (for any value of j) both give identical results. From the definition of the determinant we see that

$$\begin{aligned}
\det[A_{11}] &= A_{11} \\
\det\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} &= A_{11}A_{22} - A_{12}A_{21} \\
\det\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} &= A_{11}(A_{22}A_{33} - A_{23}A_{32}) - \\
& \quad A_{12}(A_{21}A_{33} - A_{23}A_{31}) + \\
& \quad A_{13}(A_{21}A_{32} - A_{22}A_{31})
\end{aligned} \tag{1.20}$$

Some interesting properties of determinants are

$$|AB| = |A||B| \tag{1.21}$$

assuming that A and B are square and have the same dimensions. Also,

$$|A| = \prod_{i=1}^{n} \lambda_i \tag{1.22}$$

where λ_i (the eigenvalues of A) are defined below.

The inverse of a matrix A is defined as the matrix A^{-1} such that $AA^{-1} = A^{-1}A = I$. A matrix cannot have an inverse unless it is square. Some square matrices do not have an inverse. A square matrix that does not have an inverse is called singular or invertible. In the scalar case, the only number that does not have an inverse is the number 0. But in the matrix case, there are many matrices that are singular. A matrix that does have an inverse is called nonsingular or invertible. For example, notice that

$$\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2/3 & 1/3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{1.23}$$

Therefore, the two matrices on the left side of the equation are inverses of each other. The nonsingularity of an $n \times n$ matrix A can be stated in many equivalent ways, some of which are the following [Hor85]:

- A is nonsingular.
- A^{-1} exists.
- The rank of A is equal to n.
- The rows of A are linearly independent.
- The columns of A are linearly independent.
- $|A| \neq 0$.
- $Ax = b$ has a unique solution x for all b.
- 0 is not an eigenvalue of A.

The trace of a square matrix is defined as the sum of its diagonal elements:

$$\mathrm{Tr}(A) = \sum_i A_{ii} \tag{1.24}$$

The trace of a matrix is defined only if the matrix is square. The trace of a 1×1 matrix is equal to the trace of a scalar, which is equal to the value of the scalar. One interesting property of the trace of a square matrix is

$$\mathrm{Tr}(A) = \sum_i \lambda_i \tag{1.25}$$

That is, the trace of a square matrix is equal to the sum of its eigenvalues.

Some interesting and useful characteristics of matrix products are the following:

$$\begin{aligned} (AB)^T &= B^T A^T \\ (AB)^{-1} &= B^{-1} A^{-1} \\ \mathrm{Tr}(AB) &= \mathrm{Tr}(BA) \end{aligned} \tag{1.26}$$

This assumes that the inverses exist for the inverse equation, and that the matrix dimensions are compatible so that matrix multiplication is defined. The transpose of a matrix product is equal to the product of the transposes in the opposite order. The inverse of a matrix product is equal to the product of the inverses in the opposite order. The trace of a matrix product is independent of the order in which the matrices are multiplied.

The two-norm of a column vector of real numbers, also called the Euclidean norm, is defined as follows:

$$\begin{aligned} \|x\|_2 &= \sqrt{x^T x} \\ &= \sqrt{x_1^2 + \cdots + x_n^2} \end{aligned} \tag{1.27}$$

From (1.14) we see that

$$xx^T = \begin{bmatrix} x_1^2 & \cdots & x_1 x_n \\ \vdots & \ddots & \vdots \\ x_n x_1 & \cdots & x_n^2 \end{bmatrix} \tag{1.28}$$

Taking the trace of this matrix is

$$\begin{aligned} \mathrm{Tr}(xx^T) &= x_1^2 + \cdots + x_n^2 \\ &= \|x\|_2^2 \end{aligned} \tag{1.29}$$

An $n \times n$ matrix A has n eigenvalues and n eigenvectors. The scalar λ is an eigenvalue of A, and the $n \times 1$ vector x is an eigenvector of A, if the following equation holds:

$$Ax = \lambda x \tag{1.30}$$

The eigenvalues and eigenvectors of a matrix are collectively referred to as the eigendata of the matrix.[2] An $n \times n$ matrix has exactly n eigenvalues, although

[2]Eigendata have also been referred to by many other terms over the years, including characteristic roots, latent roots and vectors, and proper numbers and vectors [Fad59].

some may be repeated. This is like saying that an nth order polynomial equation has exactly n roots, although some may be repeated. From the above definitions of eigenvalues and eigenvectors we can see that

$$\begin{aligned} Ax &= \lambda x \\ A^2x &= A\lambda x \\ &= \lambda(Ax) \\ &= \lambda(\lambda x) \\ &= \lambda^2 x \end{aligned} \tag{1.31}$$

So if A has eigendata (λ, x), then A^2 has eigendata (λ^2, x). It can be shown that A^{-1} exists if and only if none of the eigenvalues of A are equal to 0. If A is symmetric then all of its eigenvalues are real numbers.

A symmetric $n \times n$ matrix A can be characterized as either positive definite, positive semidefinite, negative definite, negative semidefinite, or indefinite. Matrix A is:

- *Positive definite* if $x^TAx > 0$ for all nonzero $n \times 1$ vectors x. This is equivalent to saying that all of the eigenvalues of A are positive real numbers. If A is positive definite, then A^{-1} is also positive definite.

- *Positive semidefinite* if $x^TAx \geq 0$ for all $n \times 1$ vectors x. This is equivalent to saying that all of the eigenvalues of A are nonnegative real numbers. Positive semidefinite matrices are sometimes called nonnegative definite.

- *Negative definite* if $x^TAx < 0$ for all nonzero $n \times 1$ vectors x. This is equivalent to saying that all of the eigenvalues of A are negative real numbers. If A is negative definite, then A^{-1} is also negative definite.

- *Negative semidefinite* if $x^TAx \leq 0$ for all $n \times 1$ vectors x. This is equivalent to saying that all of the eigenvalues of A are nonpositive real numbers. Negative semidefinite matrices are sometimes called nonpositive definite.

- *Indefinite* if it does not fit into any of the above four categories. This is equivalent to saying that some of its eigenvalues are positive and some are negative.

Some books generalize the idea of positive definiteness and negative definiteness to include nonsymmetric matrices.

The weighted two-norm of an $n \times 1$ vector x is defined as

$$||x||_Q^2 = \sqrt{x^TQx} \tag{1.32}$$

where Q is required to be an $n \times n$ positive definite matrix. The above norm is also called the Q-weighted two-norm of x. A quantity of the form x^TQx is called a quadratic in analogy to a quadratic term in a scalar equation.

The singular values σ of a matrix A are defined as

$$\begin{aligned} \sigma^2(A) &= \lambda(A^TA) \\ &= \lambda(AA^T) \end{aligned} \tag{1.33}$$

If A is an $n \times m$ matrix, then it has $\min(n,m)$ singular values. AA^T will have n eigenvalues, and A^TA will have m eigenvalues. If $n > m$ then AA^T will have the same eigenvalues as A^TA plus an additional $(n-m)$ zeros. These additional zeros are not considered to be singular values of A, because A always has $\min(n,m)$ singular values. This knowledge can help reduce effort during the computation of singular values. For example, if A is a 13×3 matrix, then it is much easier to compute the eigenvalues of the 3×3 matrix A^TA rather than the 13×13 matrix AA^T. Either computation will result in the same three singular values.

1.1.2 The matrix inversion lemma

In this section, we will derive the matrix inversion lemma, which is a tool that we will use many times in this book. It is also a tool that is frequently useful in other areas of control, estimation theory, and signal processing.

Suppose we have the partitioned matrix $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ where A and D are invertible square matrices, and the B and C matrices may or may not be square. We define E and F matrices as follows:

$$\begin{aligned} E &= D - CA^{-1}B \\ F &= A - BD^{-1}C \end{aligned} \tag{1.34}$$

Assume that E is invertible. Then we can show that

$$\begin{aligned} &\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} A^{-1} + A^{-1}BE^{-1}CA^{-1} & -A^{-1}BE^{-1} \\ -E^{-1}CA^{-1} & E^{-1} \end{bmatrix} \\ &= \begin{bmatrix} I + BE^{-1}CA^{-1} - BE^{-1}CA^{-1} & -BE^{-1} + BE^{-1} \\ CA^{-1} + CA^{-1}BE^{-1}CA^{-1} - DE^{-1}CA^{-1} & -CA^{-1}BE^{-1} + DE^{-1} \end{bmatrix} \\ &= \begin{bmatrix} I & 0 \\ CA^{-1} - (D - CA^{-1}B)E^{-1}CA^{-1} & (D - CA^{-1}B)E^{-1} \end{bmatrix} \\ &= \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} \end{aligned} \tag{1.35}$$

Now assume that F is invertible. Then we can show that

$$\begin{aligned} &\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} F^{-1} & -A^{-1}BE^{-1} \\ -D^{-1}CF^{-1} & E^{-1} \end{bmatrix} \\ &= \begin{bmatrix} AF^{-1} - BD^{-1}CF^{-1} & -BE^{-1} + BE^{-1} \\ CF^{-1} - CF^{-1} & -CA^{-1}BE^{-1} + DE^{-1} \end{bmatrix} \\ &= \begin{bmatrix} (A - BD^{-1}C)F^{-1} & 0 \\ 0 & (D - CA^{-1}B)E^{-1} \end{bmatrix} \\ &= \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} \end{aligned} \tag{1.36}$$

Equations (1.35) and (1.36) are two expressions for the inverse of $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$. Since these two expressions are inverses of the same matrix, they must be equal. We therefore conclude that the upper-left partitions of the matrices are equal, which gives

$$F^{-1} = A^{-1} + A^{-1}BE^{-1}CA^{-1} \tag{1.37}$$

Now we can use the definition of F to obtain

$$(A - BD^{-1}C)^{-1} = A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} \tag{1.38}$$

This is called the matrix inversion lemma. It is also referred to by other terms, such as the Sherman–Morrison formula, Woodbury's identity, and the modified matrices formula. One of its earliest presentations was in 1944 by William Duncan [Dun44], and similar identities were developed by Alston Householder [Hou53]. An account of its origins and variations (e.g., singular A) is given in [Hen81]. The matrix inversion lemma is often stated in slightly different but equivalent ways. For example,

$$(A + BD^{-1}C)^{-1} = A^{-1} - A^{-1}B(D + CA^{-1}B)^{-1}CA^{-1} \tag{1.39}$$

The matrix inversion lemma can sometimes be used to reduce the computational effort of matrix inversion. For instance, suppose that A is $n \times n$, B is $n \times p$, C is $p \times n$, D is $p \times p$, and $p < n$. Suppose further that we already know A^{-1}, and we want to add some quantity to A and then compute the new inverse. A straightforward computation of the new inverse would be an $n \times n$ inversion. But if the new matrix to invert can be written in the form of the left side of Equation (1.39), then we can use the right side of Equation (1.39) to compute the new inverse, and the right side of Equation (1.39) requires a $p \times p$ inversion instead of an $n \times n$ inversion (since we already know the inverse of the old A matrix).

■ **EXAMPLE 1.1**

At your investment firm, you notice that in January the New York Stock Exchange index decreased by 2%, the American Stock Exchange index increased by 1%, and the NASDAQ stock exchange index increased by 2%. As a result, investors increased their deposits by 1%. The next month, the stock exchange indices changed by −4%, 3%, and 2%, respectively, and investor deposits increased by 2%. The following month, the stock exchange indices changed by −5%, 1%, and 5%, respectively, and investor deposits increased by 2%. You suspect that investment changes y can be modeled as $y = g_1x_1 + g_2x_2 + g_3x_3$, where the x_i variables are the stock exchange index changes, and the g_i are unknown constants. In order to determine the g_i constants you need to invert the matrix

$$A = \begin{bmatrix} -2 & 1 & 2 \\ -4 & 3 & 2 \\ -5 & 1 & 5 \end{bmatrix} \tag{1.40}$$

The result is

$$\begin{aligned} A^{-1} &= \frac{1}{6}\begin{bmatrix} 13 & -3 & -4 \\ 10 & 0 & -4 \\ 11 & -3 & -2 \end{bmatrix} \\ g &= A^{-1}\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \\ &= \frac{1}{6}\begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \end{aligned} \tag{1.41}$$

This allows you to use stock exchange index changes to predict investment changes in the following month, which allows you to better schedule personnel and computer resources. However, soon afterward you find out that the NASDAQ change in the third month was actually 6% rather than 5%. This means that in order to find the g_i constants you need to invert the matrix

$$A' = \begin{bmatrix} -2 & 1 & 2 \\ -4 & 3 & 2 \\ -5 & 1 & 6 \end{bmatrix} \tag{1.42}$$

You are tired of inverting matrices and so you wonder if you can somehow use the inverse of A (which you have already calculated) to find the inverse of A'. Remembering the matrix inversion lemma, you realize that $A' = A + BD^{-1}C$, where

$$\begin{aligned} B &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T \\ C &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \\ D &= 1 \end{aligned} \tag{1.43}$$

You therefore use the matrix inversion lemma to compute

$$\begin{aligned} (A')^{-1} &= (A + BD^{-1}C)^{-1} \\ &= A^{-1} - A^{-1}B(D + CA^{-1}B)^{-1}CA^{-1} \end{aligned} \tag{1.44}$$

The $(D + CA^{-1}B)$ term that needs to be inverted in the above equation is a scalar, so its inversion is simple. This gives

$$\begin{aligned} (A')^{-1} &= \begin{bmatrix} 4.00 & 1.00 & -1.00 \\ 3.50 & -0.50 & -1.00 \\ 2.75 & -0.75 & -0.50 \end{bmatrix} \\ g &= (A')^{-1} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0.5 \\ 0.25 \end{bmatrix} \end{aligned} \tag{1.45}$$

In this example, the use of the matrix inversion lemma is not really necessary because A' (the new matrix to invert) is only 3×3. However, with larger matrices, such as 1000×1000 matrices, the computational savings that is realized by using the matrix inversion lemma could be significant.

$\nabla\nabla\nabla$

Now suppose that A, B, C, and D are matrices, with A and D being square. Then it can be seen that

$$\begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} I & -A^{-1}B \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{bmatrix} \tag{1.46}$$

This means that

$$\begin{vmatrix} A & B \\ C & D \end{vmatrix} = |A||D - CA^{-1}B| \tag{1.47}$$

Similarly, it can be shown that

$$\begin{vmatrix} A & B \\ C & D \end{vmatrix} = |D||A - BD^{-1}C| \tag{1.48}$$

These formulas are called product rules for determinants. They were first given by the Russian-born mathematician Issai Schur in a German paper [Sch17] that was reprinted in English in [Sch86].

1.1.3 Matrix calculus

In our first calculus course, we learned the mathematics of derivatives and integrals and how to apply those concepts to scalars. We can also apply the mathematics of calculus to vectors and matrices. Some aspects of matrix calculus are identical to scalar calculus, but some scalar calculus concepts need to be extended in order to derive formulas for matrix calculus.

As intuition would lead us to believe, the time derivative of a matrix is simply equal to the matrix of the time derivatives of the individual matrix elements. Also, the integral of a matrix is equal to the matrix of the integrals of the individual matrix elements. In other words, assuming that A is an $m \times n$ matrix, we have

$$\begin{aligned} \dot{A}(t) &= \begin{bmatrix} \dot{A}_{11}(t) & \cdots & \dot{A}_{1n}(t) \\ \vdots & \ddots & \vdots \\ \dot{A}_{n1}(t) & \cdots & \dot{A}_{nn}(t) \end{bmatrix} \\ \int A(t)\,dt &= \begin{bmatrix} \int A_{11}(t)\,dt & \cdots & \int A_{1n}(t)\,dt \\ \vdots & \ddots & \vdots \\ \int A_{n1}(t)\,dt & \cdots & \int A_{nn}(t)\,dt \end{bmatrix} \end{aligned} \tag{1.49}$$

Next we will compute the time derivative of the inverse of a matrix. Suppose that matrix $A(t)$, which we will denote as A, has elements that are functions of time. We know that $AA^{-1} = I$; that is, AA^{-1} is a constant matrix and therefore has a time derivative of zero. But the time derivative of AA^{-1} can be computed as

$$\frac{d}{dt}(AA^{-1}) = \dot{A}A^{-1} + A\frac{d}{dt}(A^{-1}) \tag{1.50}$$

Since this is zero, we can solve for $d(A^{-1})/dt$ as

$$\frac{d}{dt}(A^{-1}) = -A^{-1}\dot{A}A^{-1} \tag{1.51}$$

Note that for the special case of a scalar A, this reduces to the familiar equation

$$\begin{aligned} \frac{d}{dt}(1/A) &= \frac{\partial(1/A)}{\partial A}\frac{dA}{dt} \\ &= -\dot{A}/A^2 \end{aligned} \tag{1.52}$$

Now suppose that x is an $n \times 1$ vector and $f(x)$ is a scalar function of the elements of x. Then

$$\frac{\partial f}{\partial x} = \begin{bmatrix} \partial f/\partial x_1 & \cdots & \partial f/\partial x_n \end{bmatrix} \tag{1.53}$$

Even though x is a column vector, $\partial f/\partial x$ is a row vector. The converse is also true – if x is a row vector, then $\partial f/\partial x$ is a column vector. Note that some authors define this the other way around. That is, they say that if x is a column vector then $\partial f/\partial x$ is also a column vector. There is no accepted convention for the definition of the partial derivative of a scalar with respect to a vector. It does not really matter which definition we use as long as we are consistent. In this book, we will use the convention described by Equation (1.53).

Now suppose that A is an $m \times n$ matrix and $f(A)$ is a scalar. Then the partial derivative of a scalar with respect to a matrix can be computed as follows:

$$\frac{\partial f}{\partial A} = \begin{bmatrix} \partial f/\partial A_{11} & \cdots & \partial f/\partial A_{1n} \\ \vdots & \ddots & \vdots \\ \partial f/\partial A_{m1} & \cdots & \partial f/\partial A_{mn} \end{bmatrix} \tag{1.54}$$

With these definitions we can compute the partial derivative of the dot product of two vectors. Suppose x and y are n-element column vectors. Then

$$\begin{aligned} x^T y &= x_1 y_1 + \cdots + x_n y_n \\ \frac{\partial(x^T y)}{\partial x} &= \begin{bmatrix} \partial(x^T y)/\partial x_1 & \cdots & \partial(x^T y)/\partial x_n \end{bmatrix} \\ &= \begin{bmatrix} y_1 & \cdots & y_n \end{bmatrix} \\ &= y^T \end{aligned} \tag{1.55}$$

Likewise, we can obtain

$$\frac{\partial(x^T y)}{\partial y} = x^T \tag{1.56}$$

Now we will compute the partial derivative of a quadratic with respect to a vector. First write the quadratic as follows:

$$\begin{aligned} x^T A x &= \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{n1} & \cdots & A_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\ &= \begin{bmatrix} \sum_i x_i A_{i1} & \cdots & \sum_i x_i A_{in} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\ &= \sum_{i,j} x_i x_j A_{ij} \end{aligned} \tag{1.57}$$

Now take the partial derivative of the quadratic as follows:

$$\begin{aligned} \frac{\partial(x^T A x)}{\partial x} &= \begin{bmatrix} \partial(x^T A x)/\partial x_1 & \cdots & \partial(x^T A x)/\partial x_n \end{bmatrix} \\ &= \begin{bmatrix} \sum_j x_j A_{1j} + \sum_i x_i A_{i1} & \cdots & \sum_j x_j A_{nj} + \sum_i x_i A_{in} \end{bmatrix} \\ &= \begin{bmatrix} \sum_j x_j A_{1j} & \cdots & \sum_j x_j A_{nj} \end{bmatrix} + \begin{bmatrix} \sum_i x_i A_{i1} & \cdots & \sum_i x_i A_{in} \end{bmatrix} \\ &= x^T A^T + x^T A \end{aligned} \tag{1.58}$$

If A is symmetric, as it often is in quadratic expressions, then $A = A^T$ and the above expression simplifies to

$$\frac{\partial(x^TAx)}{\partial x} = 2x^TA \qquad \text{if } A = A^T \tag{1.59}$$

Next we define the partial derivative of a vector with respect to another vector. Suppose $g(x) = \begin{bmatrix} g_1(x) \\ \vdots \\ g_m(x) \end{bmatrix}$ and $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$. Then

$$\frac{\partial g}{\partial x} = \begin{bmatrix} \partial g_1/\partial x_1 & \cdots & \partial g_1/\partial x_n \\ \vdots & & \vdots \\ \partial g_m/\partial x_1 & \cdots & \partial g_m \partial x_n \end{bmatrix} \tag{1.60}$$

If either $g(x)$ or x is transposed, then the partial derivative is also transposed.

$$\begin{aligned} \frac{\partial g^T}{\partial x} &= \left(\frac{\partial g}{\partial x}\right)^T \\ \frac{\partial g}{\partial x^T} &= \left(\frac{\partial g}{\partial x}\right)^T \\ \frac{\partial g^T}{\partial x^T} &= \frac{\partial g}{\partial x} \end{aligned} \tag{1.61}$$

With these definitions, the following important equalities can be derived. Suppose A is an $m \times n$ matrix and x is an $n \times 1$ vector. Then

$$\begin{aligned} \frac{\partial(Ax)}{\partial x} &= A \\ \frac{\partial(x^TA)}{\partial x} &= A \end{aligned} \tag{1.62}$$

Now we suppose that A is an $m \times n$ matrix, B is an $n \times n$ matrix, and we want to compute the partial derivative of $\mathrm{Tr}(ABA^T)$ with respect to A. First compute ABA^T as follows:

$$\begin{aligned} ABA^T &= \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix} \begin{bmatrix} B_{11} & \cdots & B_{1n} \\ \vdots & \ddots & \vdots \\ B_{n1} & \cdots & B_{nn} \end{bmatrix} \begin{bmatrix} A_{11} & \cdots & A_{m1} \\ \vdots & \ddots & \vdots \\ A_{1n} & \cdots & A_{nn} \end{bmatrix} \\ &= \begin{bmatrix} \sum_{j,k} A_{1k}B_{kj}A_{1j} & \cdots & \sum_{j,k} A_{1k}B_{kj}A_{mj} \\ \vdots & & \vdots \\ \sum_{j,k} A_{mk}B_{kj}A_{1j} & \cdots & \sum_{j,k} A_{mk}B_{kj}A_{mj} \end{bmatrix} \end{aligned} \tag{1.63}$$

From this we see that the trace of ABA^T is given as

$$\mathrm{Tr}(ABA^T) = \sum_{i,j,k} A_{ik}B_{kj}A_{ij} \tag{1.64}$$

Its partial derivative with respect to A can be computed as

$$
\begin{aligned}
\frac{\partial \mathrm{Tr}(ABA^T)}{\partial A} &= \begin{bmatrix} \partial \mathrm{Tr}(ABA^T)/\partial A_{11} & \cdots & \partial \mathrm{Tr}(ABA^T)/\partial A_{1n} \\ \vdots & & \vdots \\ \partial \mathrm{Tr}(ABA^T)/\partial A_{m1} & \cdots & \partial \mathrm{Tr}(ABA^T)/\partial A_{mn} \end{bmatrix} \\
&= \begin{bmatrix} \sum_j A_{1j}B_{1j} + \sum_k A_{1k}B_{k1} & \cdots & \sum_j A_{1j}B_{nj} + \sum_k A_{1k}B_{kn} \\ \vdots & \ddots & \vdots \\ \sum_j A_{mj}B_{1j} + \sum_k A_{mk}B_{k1} & \cdots & \sum_j A_{mj}B_{nj} + \sum_k A_{mk}B_{kn} \end{bmatrix} \\
&= \begin{bmatrix} \sum_j A_{1j}B_{1j} & \cdots & \sum_j A_{1j}B_{nj} \\ \vdots & & \vdots \\ \sum_j A_{mj}B_{1j} & \cdots & \sum_j A_{mj}B_{nj} \end{bmatrix} + \\
&\quad \begin{bmatrix} \sum_k A_{1k}B_{k1} & \cdots & \sum_k A_{1k}B_{kn} \\ \vdots & & \vdots \\ \sum_k A_{mk}B_{k1} & \cdots & \sum_k A_{mk}B_{kn} \end{bmatrix} \\
&= AB^T + AB
\end{aligned} \tag{1.65}
$$

If B is symmetric, as it often is in partial derivatives of the form above, then this can be simplified to

$$
\frac{\partial \mathrm{Tr}(ABA^T)}{\partial A} = 2AB \qquad \text{if } B = B^T \tag{1.66}
$$

A number of additional interesting results related to matrix calculus can be found in [Ske98, Appendix B].

1.1.4 The history of matrices

This section is a brief diversion to present some of the history of matrix theory. Much of the information in this section is taken from [OCo96].

The use of matrices can be found as far back as the fourth century BC. We see in ancient clay tablets that the Babylonians studied problems that led to simultaneous linear equations. For example, a tablet dating from about 300 BC contains the following problem: "There are two fields whose total area is 1800 units. One produces grain at the rate of 2/3 of a bushel per unit while the other produces grain at the rate of 1/2 a bushel per unit. If the total yield is 1100 bushels, what is the size of each field?"

Later, the Chinese came even closer to the use of matrices. In [She99] (originally published between 200 BC and 100 AD) we see the following problem: "There are three types of corn, of which three bundles of the first, two of the second, and one of the third make 39 measures. Two of the first, three of the second, and one of the third make 34 measures. And one of the first, two of the second and three of the third make 26 measures. How many measures of corn are contained in one bundle of each type?" At that point, the ancient Chinese essentially use Gaussian elimination (which was not well known until the 19th century) to solve the problem.

In spite of this very early beginning, it was not until the end of the 17th century that serious investigation of matrix algebra began. In 1683, the Japanese